

1. Problem Statement

In traditional retail, bargaining provides buyers with a sense of control, fairness, and engagement. However, e-commerce platforms typically rely on static discounts, coupon codes, or seasonal sales, which lack personalization and fail to replicate the interactive nature of real-world negotiations. Buyers often feel dissatisfied when their requests for discounts are ignored, while sellers cannot afford to manually handle negotiations at scale.

The absence of a dynamic negotiation system in online shopping creates a gap between **buyer expectations** and **seller feasibility**. There is a pressing need for an intelligent, automated solution that can negotiate within defined business constraints, adapt to customer sentiment, and simultaneously enhance user engagement without compromising profitability.

2. Objectives

1. **Develop an AI-driven negotiation assistant** that interacts with customers in natural language during checkout.
2. **Integrate a rule-based pricing engine** to ensure discounts and offers remain within predefined profit margins.
3. **Implement sentiment analysis** (using TextBlob or VADER) to adapt responses based on customer tone and mood.
4. **Introduce gamification elements** such as “Bargain Points,” loyalty scores, and badges to make negotiations engaging and encourage repeat visits.
5. **Maintain personalized offer history** so users can track their past negotiations and rewards, fostering transparency and trust.
6. **Demonstrate scalability and feasibility** by designing a modular system architecture that can be extended for real-world e-commerce integration.

3. Literature Survey

Artificial Intelligence (AI) and Natural Language Processing (NLP) have significantly transformed the way businesses interact with customers, especially in e-commerce and automated negotiation systems. Several studies have explored the integration of AI in business negotiation frameworks to automate pricing and enhance decision-making. *AI-Powered Negotiations: Opportunities, Challenges, and the Future of Business Strategy* (2023) and *Price Negotiation Chatbot Using AI-Based Ensemble Learning* (IJIRT, 2024) emphasized that AI-driven negotiation systems enable dynamic pricing, improved scalability, and efficient handling of large-scale customer interactions. Sentiment analysis further enhances these systems by allowing adaptive and emotionally aware communication. *Rajesh Paul et al. (2024)* demonstrated that sentiment analysis tools such as VADER and TextBlob effectively interpret customer tone and attitude, enabling systems to tailor responses to user emotions. Similarly, *Emely Bury (2024)* and *BrimLabs AI (2023)* discussed the role of gamification in enhancing user engagement and loyalty through mechanisms like points, badges, and leaderboards, where AI personalizes rewards to sustain user motivation. Furthermore, *P. Vijayaragavan (2024)* proposed a hybrid sentiment-based model for personalized recommendations, improving satisfaction through customized interactions and transparent offer generation.

Despite these advancements, a clear research gap exists in integrating all three elements—AI negotiation, sentiment analysis, and gamification—into a single cohesive platform. Existing systems either focus on automated negotiation or sentiment adaptation but rarely combine emotional intelligence with dynamic, reward-driven interaction. The proposed AI-Driven Negotiation Assistant addresses this gap by merging real-time bargaining, mood-based adaptation, and gamified loyalty systems to create a human-like, engaging, and profit-optimized negotiation experience for modern e-commerce platforms.

4. Design Details

4.1 UML Class Diagram

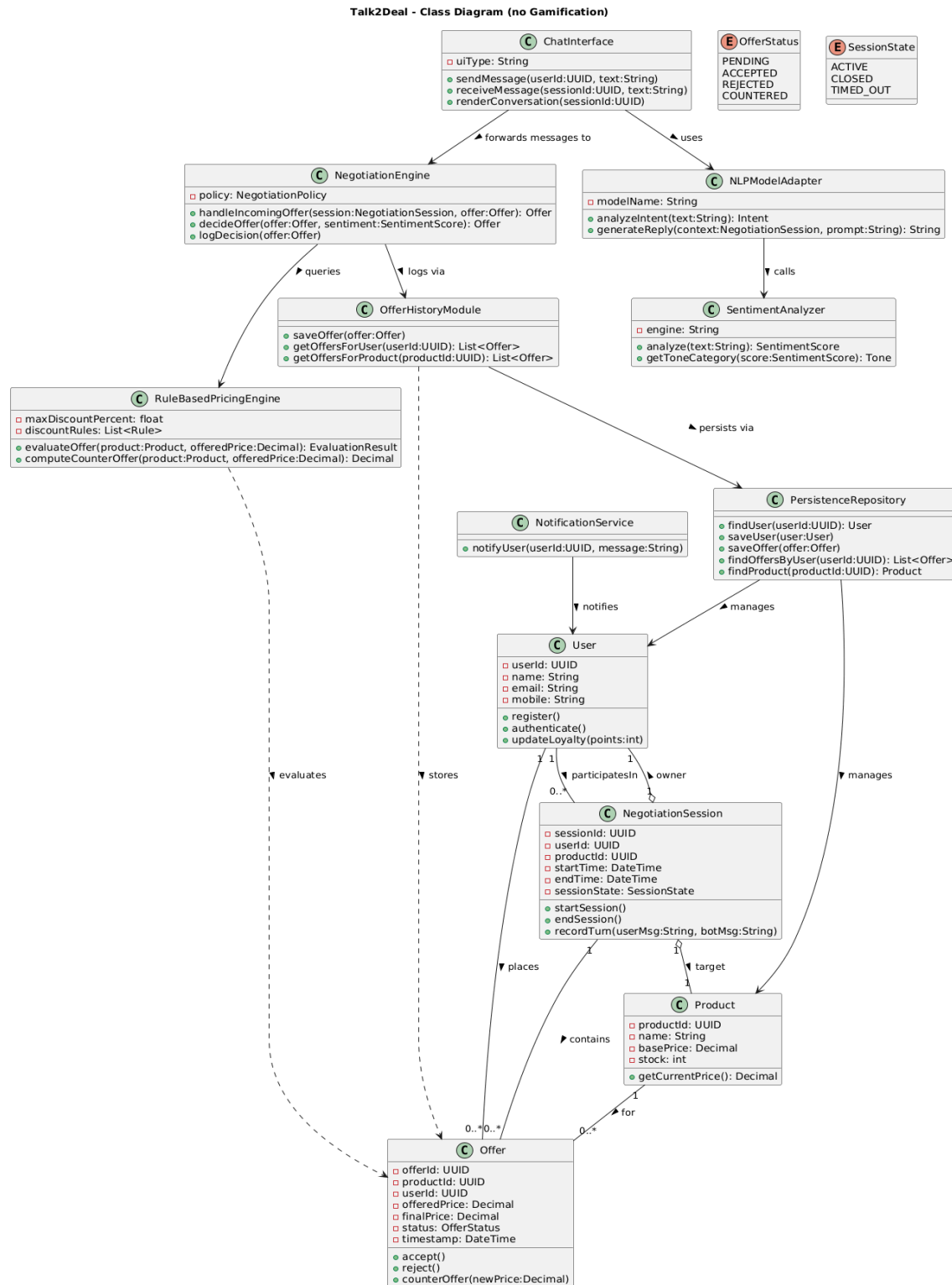


Fig.4.1 - Shows system classes, attributes, and relationships for AI-driven negotiation workflow.

4.2 UML Sequence Diagram

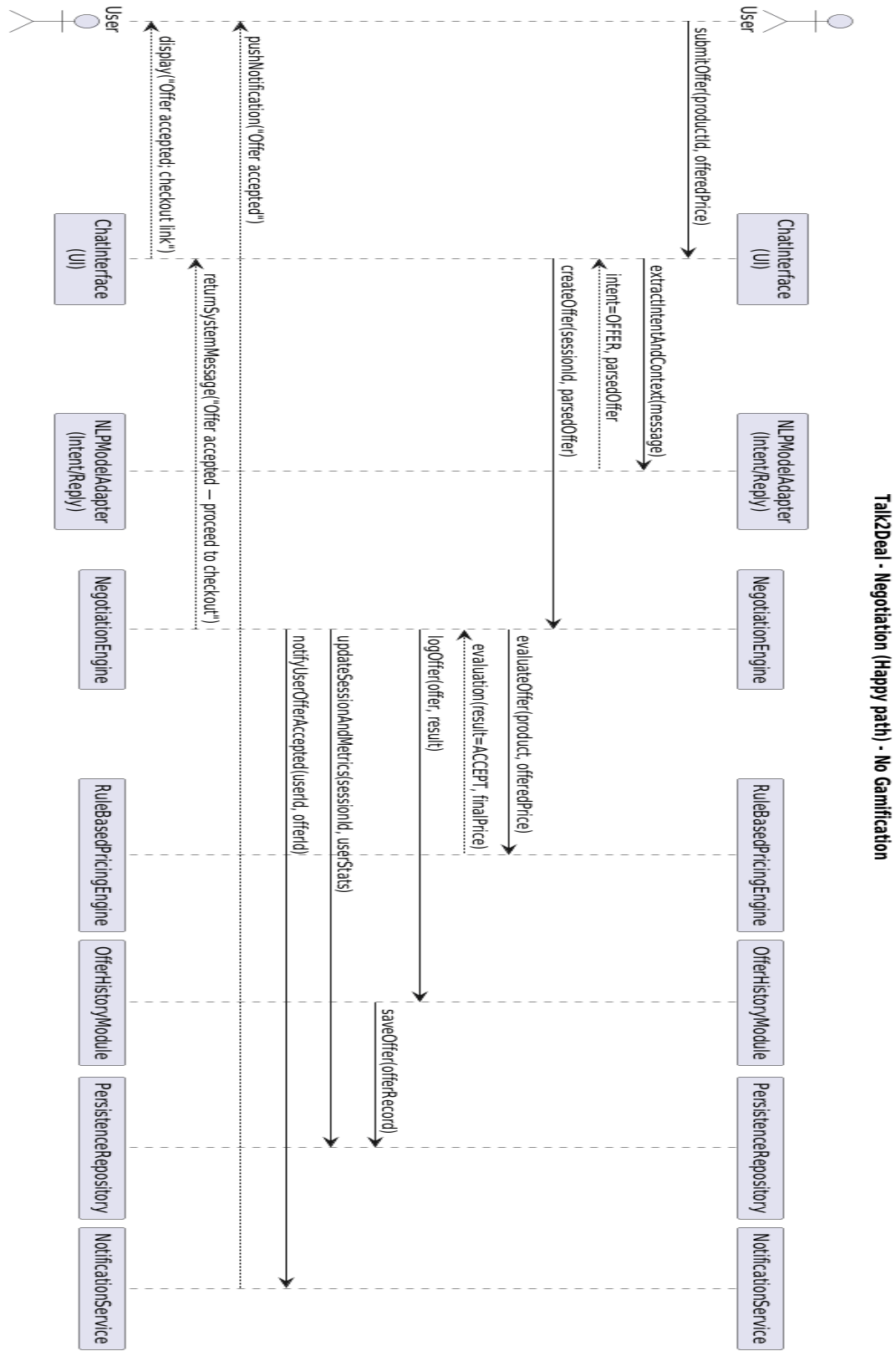


Fig 4.2 - Illustrates message flow between user and system during negotiation process.

5. Requirements Analysis & Feasibility

5.1 Requirements Analysis

The Talk2Deal platform integrates natural-language negotiation, dynamic pricing logic, user authentication, and cloud-based data management. The system must intelligently process user inputs, interpret sentiment and discount intent, generate counteroffers, and maintain consistent negotiation histories. These requirements demand:

- A robust backend capable of real-time processing
- A conversational AI model capable of accurate interpretation
- Scalable cloud storage for user and negotiation data
- A user interface that simplifies negotiation interactions
- Administrative tools for system oversight and product management

The outcome of the requirements study confirms that the system can be structured into distinct, well-defined modules:

- LLM Processing
- Pricing Engine
- Data Access Layer
- API Layer (FastAPI)
- Client UI
- Admin Dashboard

Each module supports a specific operational responsibility and contributes to the overall negotiation workflow.

5.2 Feasibility Analysis

A detailed feasibility assessment was conducted to evaluate the practicality of implementing the system under the current technological, operational, and financial constraints.

5.2.1 Technical Feasibility

The technological components selected for the platform are modern, reliable, and highly compatible with the system's intended functionality.

- **Large Language Model (Gemini API)**
 - Provides advanced conversational understanding, sentiment analysis, and intent extraction.
 - Supports real-time, context-aware negotiation responses.
 - Fully cloud-hosted, eliminating the need for onsite AI infrastructure.
- **Firestore NoSQL Cloud Database**
 - Scalable, fast, and optimized for real-time applications.
 - Allows seamless storage of negotiation logs, user profiles, and state updates.

- Ensures consistency and reliability across user sessions.
- **FastAPI Backend Framework**
 - High-performance asynchronous API engine with excellent response times.
 - Ideal for handling negotiation flows that require quick LLM API calls and database interactions.
 - Modular structure simplifies maintenance and future system enhancements.
- **Browser-Based Client Interface**
 - Accessible across multiple devices without installation.
 - Lightweight and easy to integrate with backend APIs.

5.2.2 Operational Feasibility

The system is designed to be intuitive and convenient for both customers and administrators.

- The negotiation experience resembles natural conversation, ensuring minimal learning curve.
- Customers can select products, send messages, and receive responses instantly, enhancing engagement.
- Admin users can monitor negotiations, manage products, and review system activity through a dedicated dashboard.
- Cloud hosting ensures high availability without complex system management.

5.2.3 Economic Feasibility

The financial assessment supports the viability of the system.

- **Pay-as-you-go AI Billing (Gemini API)**
 - Cost is based solely on actual usage.
 - Scales according to business activity without upfront investment.
- **Firestore Pricing Model**
 - Charges for storage and read/write operations only.
 - Eliminates traditional database hosting costs.
- **Low Deployment Costs**
 - FastAPI can be deployed on affordable cloud services (Google Cloud Run, Render, Railway).
 - No physical servers, networking hardware, or infrastructure required.

6. Functional and Non-Functional Requirements

This chapter outlines the essential requirements that define the behavior, quality, and constraints of the Talk2Deal AI-Driven Negotiation System. These requirements ensure that the system performs its intended functions reliably while meeting performance, usability, and security expectations.

6.1 Functional Requirements

Functional requirements specify the features and operations the system must support.

6.1.1 User Registration and Authentication

1. The system shall allow new users to register using a unique user ID and password.
2. The system shall validate login credentials and authenticate users securely.
3. The system shall prevent unauthorized access to negotiation and profile data.

6.1.2 Product Viewing and Selection

1. The system shall display a list of available products retrieved from Firestore.
2. The system shall provide product details such as name and base price.
3. The user shall select a product before initiating negotiation.

6.1.3 Negotiation Process

1. The system shall allow users to initiate a negotiation session for a selected product.
2. The system shall capture and send user messages to the backend for analysis.
3. The system shall process user inputs through the LLMService to extract sentiment and requested discount.
4. The system shall compute a counteroffer using the PricingEngine based on sentiment, negotiation history, and product rules.
5. The system shall return AI-generated messages that clearly communicate the offer or negotiation status.
6. The system shall detect when negotiation reaches an acceptance or termination state.

6.1.4 Gamification and Rewards

1. The system shall track user Bargain Points based on negotiation outcomes.
2. The system shall update and store accumulated points in the user's profile.
3. The system shall reflect reward points in both negotiation flow and client interface.

6.1.5 Data Storage and Logging

1. The system shall store negotiation history in Firestore for each user-product pair.
2. The system shall maintain the most recent offer and relevant negotiation metrics.
3. The system shall log each turn with user message, AI decision, and extracted metadata.

6.1.6 Admin Functionalities

1. The admin dashboard shall allow managing product catalogs, including adding, modifying, and deleting products.

2. The dashboard shall display negotiation logs for monitoring user activity.
3. The admin shall be able to view aggregated insights and negotiation summaries.

6.2 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance expectations, and constraints of the system.

6.2.1 Performance Requirements

1. The system shall generate negotiation responses within **1–2 seconds** under normal network conditions.
2. The backend shall handle concurrent negotiation sessions efficiently.
3. The Firestore database shall respond to read/write operations with minimal latency.

6.2.2 Scalability Requirements

1. The system shall support an increasing number of users without degradation in performance.
2. Cloud services (Firestore, Gemini API) shall scale automatically with demand.
3. The architecture shall support modular expansion without requiring restructuring.

6.2.3 Security Requirements

1. User passwords shall be securely stored using hashing techniques.
2. API keys (Gemini, Firebase) shall not be exposed on the client side.
3. Unauthorized users shall be prevented from accessing restricted admin functionalities.
4. All communication between client and server shall follow secure protocols.

6.2.4 Usability Requirements

1. The client interface shall be intuitive and easy to navigate for users with no technical background.
2. The negotiation chat interface shall mimic real-time conversation to enhance user engagement.
3. The admin dashboard shall be simple and logically structured for non-technical personnel.

6.2.5 Reliability Requirements

1. The system shall ensure accurate retrieval and storage of negotiation history.
2. The backend shall continue functioning even if individual components (e.g., LLM API) temporarily fail by using fallback responses.
3. The system shall maintain consistent performance across different devices and browsers.

6.2.6 Maintainability Requirements

1. The codebase shall follow a modular design (LLM, pricing engine, database access, routing).
2. Developers shall be able to update or replace individual modules without affecting the entire system.

3. Logging mechanisms shall assist in debugging and system monitoring.

6.2.7 Availability Requirements

1. The system shall maintain **high availability**, leveraging cloud infrastructure to ensure minimal downtime.
2. User data shall remain accessible even during backend restarts.

7. Code / Module / Working Prototype Developed

The Talk2Deal platform has been fully implemented as a working prototype integrating artificial intelligence, dynamic pricing, gamification, and real-time negotiation capabilities. The development process followed a modular architecture, ensuring clear separation of responsibilities, maintainability, and scalable system behavior. The prototype includes both **backend services**, **AI-driven negotiation logic**, and a **browser-based client interface**, along with an **admin dashboard** for monitoring and management.

7.1 System Modules Developed

The system has been implemented through a set of interconnected modules, each responsible for a key functionality within the platform.

7.1.1 LLM Processing Module (LLMService)

This module integrates with the **Google Gemini API** to provide the AI capabilities of the system.

Key functionalities include:

- Interpreting user messages to extract sentiment and requested discount.
- Generating contextual, human-like negotiation responses.
- Handling fallback logic when API responses fail.

The service encapsulates all AI-related logic, ensuring abstraction from the rest of the system.

7.1.2 Pricing Engine Module

The **PricingEngine** implements rule-based dynamic pricing strategies.

It analyzes sentiment, negotiation history, and user intent to compute counteroffers.

Responsibilities:

- Calculate discount offers within defined business constraints.
- Ensure discounts do not exceed requested values or maximum allowable thresholds.
- Provide a structured decision object for the AI layer (COUNTER, ACCEPT, PERK).

7.1.3 Data Access Layer (DataAccess)

This module handles all interactions with the **Firestore** database.

Features:

- Storing negotiation states
- Saving user registration details
- Recording negotiation logs
- Retrieving and updating user profiles

By centralizing database operations, the system ensures consistency and simplifies future maintenance.

7.1.4 FastAPI Backend (main.py)

The backend exposes RESTful API endpoints for:

- User Registration
- User Login
- Product Listing
- Negotiation Processing (/negotiate)
- Client and Admin View Rendering

It orchestrates communication between the frontend, LLM service, pricing engine, and Firestore.

7.1.5 Client Interface (client.html)

A fully functional **browser-based negotiation interface** was developed using HTML, CSS, and JavaScript.

Features:

- Login-based access
- Product selection
- Interactive negotiation chat window
- Real-time display of AI responses and offers
- Display of Bargain Points and final negotiated price

This interface provides a smooth user experience that simulates actual negotiations.

7.1.6 Admin Dashboard (streamlit_admin.py)

A dedicated administrative interface built using **Streamlit**.

Capabilities:

- View product catalogs
- Add or modify product pricing rules
- Monitor negotiation logs in real time
- Oversee user profiles and Bargain Points

This dashboard supports system management and validation.

7.2 Code Implementation Overview

The prototype is implemented in Python using FastAPI, Firestore SDK, and Google Generative AI SDK.

HTML, CSS, and JavaScript are used for the client-facing interface, while the admin dashboard uses Streamlit.

Major technologies used:

- **Programming Language:** Python 3.x

- **Framework:** FastAPI
- **AI Model:** Google Gemini API
- **Database:** Firestore (Firebase)
- **Frontend:** HTML, CSS, JavaScript
- **Admin UI:** Streamlit
- **Deployment:** Local environment with venv, API-based cloud services

All code follows modular patterns and adheres to best practices for readability, maintainability, and scalability.

7.3 Working Prototype Demonstration

The completed prototype demonstrates:

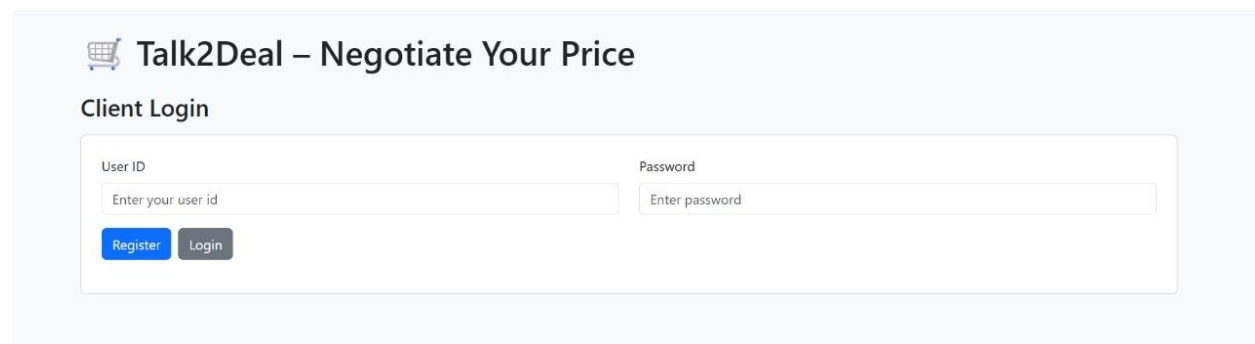


Fig 7.3.1 – login and register page of the project

Users can register new profiles, login securely, and maintain persistent accounts in Firestore.

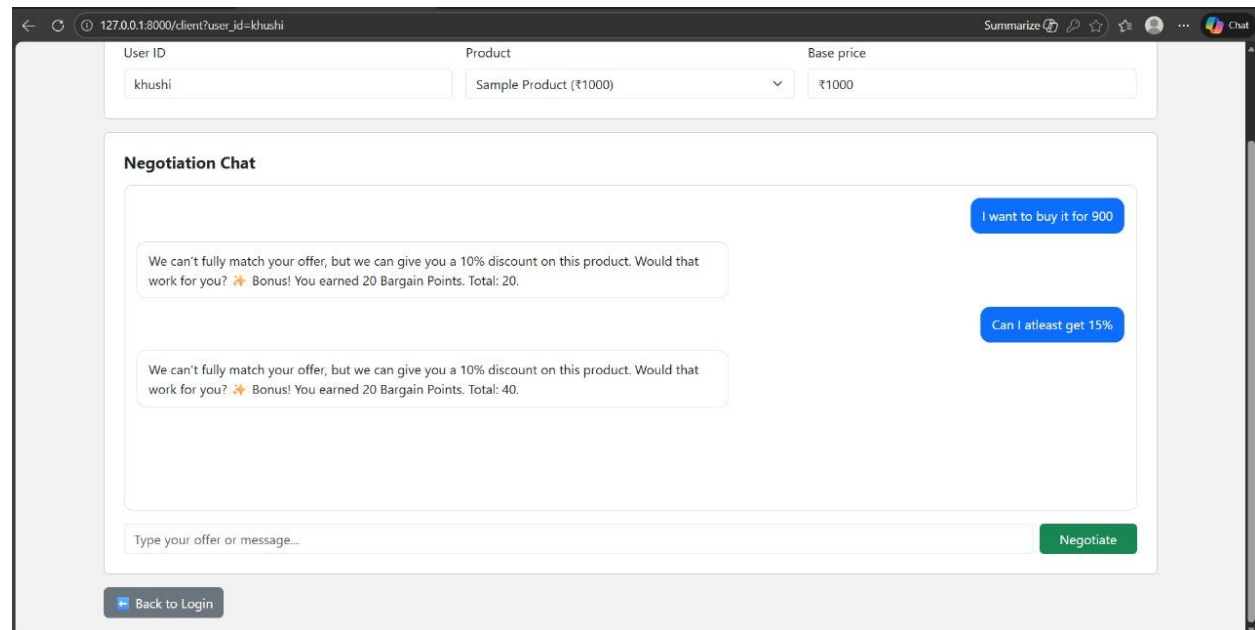


Fig 7.3.2 – Client page where negotiations take place

The system supports natural-language negotiation, with AI generating personalized counteroffers based on:

- Sentiment
- Requested discount
- Negotiation history
- Product pricing rules

Users earn Bargain Points for successful outcomes, increasing engagement.

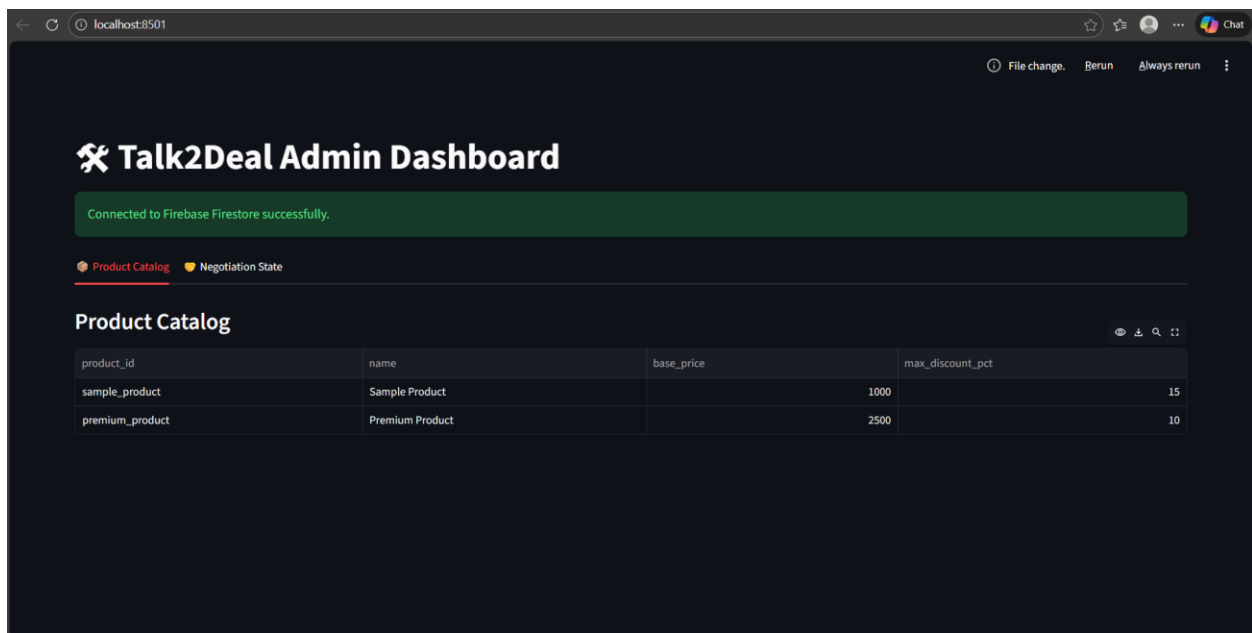


Fig 7.3.3 – Product catalog page for the admin

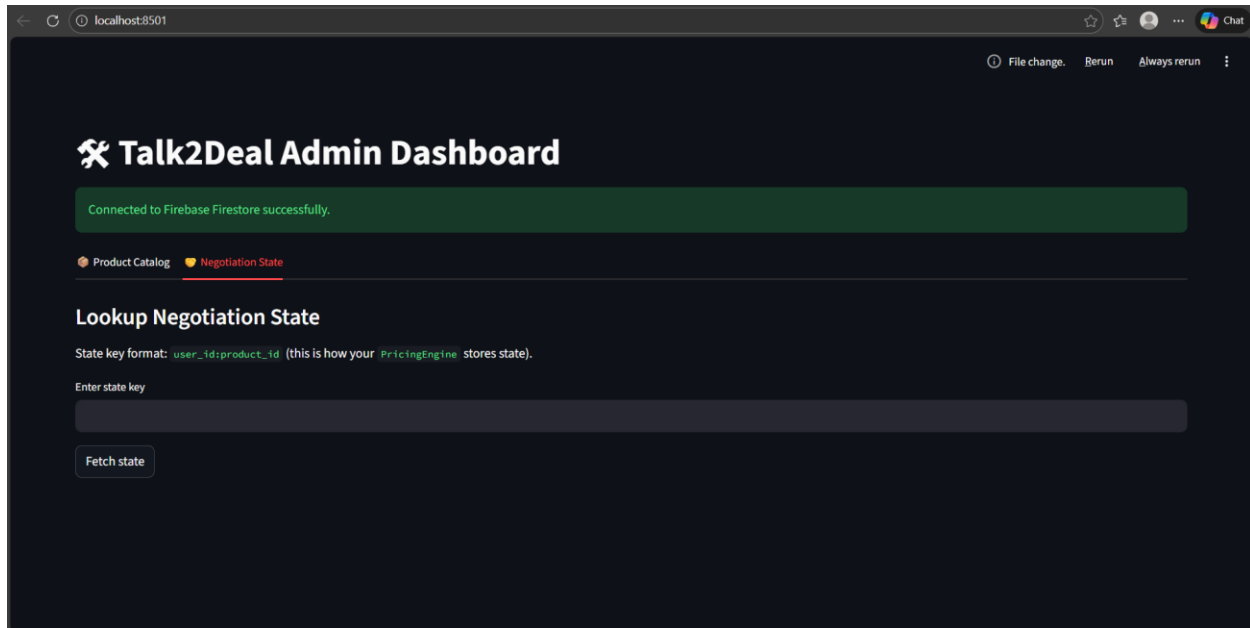


Fig 7.3.4 – Negotiation status depending on the user negotiation and product catalog.

Admin users can inspect negotiation logs, monitor activity, and manage products through Streamlit.

7.4 Summary of Prototype Achievements

The working prototype successfully integrates:

- AI-based sentiment and negotiation interpretation
- Dynamic pricing calculations
- A functional user interface
- A complete authentication and product workflow
- Real-time communication across all modules

This validates the system's design and establishes a strong foundation for future enhancements, including advanced ML-based pricing models or full e-commerce integration.

8. System Design

The System Design phase defines the architecture, structural components, data flow, interactions, and functional organization of the Talk2Deal AI-Driven Negotiation Platform. The design integrates an AI negotiation engine, cloud database, user interface, and administrative tools into a cohesive, modular, and scalable system. This chapter outlines the architectural decisions, system components, data models, and workflow processes implemented in the working prototype.

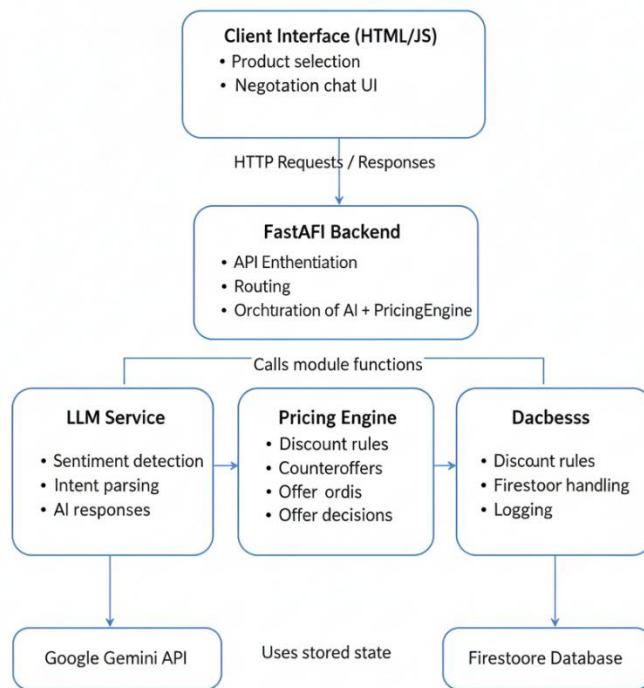
8.1 Architectural Overview

The system is built using a **multi-layered architecture** consisting of:

1. **Presentation Layer** – Client Interface & Admin Dashboard
2. **Application Layer** – FastAPI Backend
3. **AI Layer** – LLMService powered by Google Gemini
4. **Pricing Layer** – PricingEngine module
5. **Data Layer** – Firestore NoSQL Cloud Database

This layered structure ensures clear separation of concerns, maintainability, and improved performance.

8.2 System Architecture Diagram (Logical View)



8.3 Component Design

8.3.1 LLM Service (LLMService)

The LLMService module acts as the intelligence layer responsible for:

- Interpreting user messages
- Extracting sentiment and requested discount
- Generating natural, varied negotiation responses

Design Features:

- Calls Gemini generateContent() API
- Prompts are structured to ensure consistent JSON extraction
- Includes fallback handling for API failures
- Maintains modularity independent of other business logic

8.3.2 PricingEngine

The PricingEngine is responsible for computing the appropriate business decision in each negotiation turn.

Responsibilities:

- Use sentiment analysis results
- Use requested discount
- Retrieve last offer from Firestore
- Enforce business constraints (max discount, upper bounds)
- Return a structured PricingDecision object for the AI

Example decision types:

- COUNTER – propose a discount
- ACCEPT – finalize the negotiation
- PERK – offer alternative incentives

This module ensures the AI remains consistent with real business rules.

8.3.3 DataAccess Layer

The DataAccess module provides an abstraction over the Firestore database.

Key Responsibilities:

- Fetch and update user profiles
- Store negotiation state (last_offer_pct)
- Log negotiation interactions
- Manage product catalogs (read/write)

This ensures the backend remains database-agnostic and maintainable.

8.3.4 FastAPI Backend

The FastAPI application serves as the core coordinator of the system.

Features:

- Contains REST endpoints like /api/login, /api/register, /api/products, and /negotiate
- Validates user input
- Orchestrates calls between LLMService and PricingEngine
- Returns structured responses to the frontend
- Renders HTML templates (login page, negotiation page)

Its asynchronous capabilities make it ideal for AI-driven applications requiring fast responses.

8.3.5 Client Interface

The frontend is a lightweight HTML/JavaScript UI featuring:

- Login page
- Product selection dropdown
- Negotiation chat interface
- Display of real-time AI responses

- Display of final negotiated price and Bargain Points

The UI communicates with the backend using AJAX/Fetch API calls.

8.3.6 Admin Dashboard (Streamlit)

The admin UI is developed using **Streamlit** and includes:

- Product catalog management
- Negotiation activity logs
- User profile and Bargain Points monitoring
- Debugging tools for developers

Provides a simple yet powerful interface without frontend complexity.

8.4 Data Design

8.4.1 Database Schema (Firestore)

While Firestore is schema-less, the system maintains consistent collection structures:

Users Collection

user_id: string
password_hash: string
bargain_points: number
created_at: timestamp

Products Collection

product_id: string
name: string
base_price: number
max_discount_pct: number

NegotiationState Collection

state_key: user_id:product_id
last_offer_pct: number
history: list (optional)

NegotiationLogs Collection

state_key: string
user_message: string
ai_message: string
metadata: object (llm_output, pricing details)

8.5 Process Flow Design

8.5.1 Negotiation Workflow

1. User selects a product.
2. User sends a message.
3. Backend receives message and calls `analyze_user_input()`.
4. LLM returns sentiment + requested discount.
5. PricingEngine receives LLM output + product rules.
6. PricingEngine returns a structured pricing decision.
7. LLMService generates a conversational response.
8. Firestore updates negotiation state.
9. Response delivered to user UI.

9. Proposed Viable Solution

The proposed solution for Talk2Deal is an AI-driven, automated negotiation platform designed to simulate real human bargaining in an e-commerce environment. It leverages advanced natural language processing (NLP), dynamic pricing strategies, and cloud-based data management to deliver a personalized, scalable, and efficient negotiation system. This solution addresses the challenges of static pricing, limited human support, and low customer engagement by enabling real-time, data-informed negotiation experiences.

9.1 Solution Overview

The solution integrates four major technological pillars:

1. **AI Conversation Engine** using Google Gemini
2. **Dynamic Pricing Engine** that adapts to user sentiment and negotiation patterns
3. **Cloud Firestore Database** for scalable and structured data management
4. **Frontend Client Interface & Admin Dashboard** for seamless interaction and system monitoring

Together, these components create a holistic platform that modernizes negotiation processes in digital commerce.

9.2 How the Proposed Solution Addresses the Problem

Traditional e-commerce platforms lack personalized pricing and negotiation capabilities. The proposed system solves this by enabling:

- **Interactive Negotiation:** Customers bargain in natural language, similar to offline purchasing.
- **Automated Decision-Making:** The AI interprets user intent and sentiment, enabling real-time decisions without human intervention.
- **Dynamic Discounting:** The pricing engine offers discounts, perks, or acceptance based on business rules and user input.
- **Engagement Enhancement:** Gamification (Bargain Points) encourages repeated interactions and higher customer retention.
- **Scalable Data Handling:** Firestore ensures persistent storage of negotiation histories, user profiles, and product metadata.

This results in a more immersive, personalized, and flexible shopping experience.

9.3 Core Components of the Proposed Solution

9.3.1 AI-Powered Negotiation Layer

The system employs **Google Gemini** to analyze messages and generate contextual responses.

- Detects **sentiment** (positive, neutral, negative)
- Extracts **requested discount**
- Personalizes responses
- Ensures conversation remains concise and human-like

This transforms a basic chatbot into a **negotiation-aware AI agent**.

9.3.2 Dynamic Pricing Engine

A dedicated PricingEngine evaluates:

- Sentiment analysis results
- User bargaining patterns
- Maximum allowable discounts
- Product-specific business constraints

Based on these inputs, it outputs a decision from:

- **COUNTER** (discount offer)
- **ACCEPT** (finalize the deal)
- **PERK** (non-monetary reward)

This introduces **flexible, data-driven pricing** rather than static pricing models.

9.3.3 Cloud-Based Data Storage

Firestore is used to manage:

- User accounts (ID, password hash, points)
- Product data and pricing rules
- Negotiation logs per user-product pair
- Historical conversation context

The cloud-based architecture ensures:

- High availability
- Automatic scaling
- Easy integration across modules

9.3.4 User and Admin Interfaces

The solution includes two front-facing interfaces:

Client Interface (Negotiation Portal)

- Allows users to register/login
- Select products
- Engage in AI-driven negotiations
- View current price and Bargain Points

Admin Interface (Streamlit Dashboard)

- Manage product catalog
- View system activity

- Analyze negotiation logs
- Perform administrative updates

This creates a comprehensive ecosystem for both customers and administrators.

9.4 Advantages of the Proposed Solution

The proposed system delivers multiple benefits:

Customer Benefits

- Personalized shopping experience
- Opportunity to negotiate like in traditional marketplaces
- Transparent offer decisions
- Gamified engagement through points

Business Benefits

- Automation reduces human workload
- Data-driven insights into customer behaviour
- Dynamic pricing enhances conversion rates
- Scalable architecture supports high traffic

Technical Benefits

- Modular structure simplifies maintenance
- Cloud-native components improve performance
- AI integration ensures high-quality natural interactions

9.5 Viability Assessment

The solution is considered viable based on:

Technical Viability

- All APIs, frameworks, and databases are stable, widely used, and well-documented
- Modular architecture allows easy scalability and upgrades

Operational Viability

- Easy-to-use interfaces for customers and administrators
- Automation ensures efficiency and consistent outcomes

Economic Viability

- Minimal upfront investment
- Pay-as-you-go billing for Google APIs and Firebase
- Suitable for academic, prototype, and commercial use

10. Conclusion and Future Scope

10.1 Conclusion

The Talk2Deal AI-Driven Negotiation System successfully demonstrates the integration of artificial intelligence, dynamic pricing strategies, and cloud-based data management to create an interactive and intelligent negotiation experience for e-commerce users. The platform addresses key limitations of traditional online shopping systems—such as static pricing, lack of personalization, and limited user engagement—by offering a flexible and automated negotiation mechanism powered by Google Gemini.

The system efficiently analyzes user sentiment, interprets discount intentions, and generates appropriate counteroffers through its modular architecture consisting of the LLMService, PricingEngine, DataAccess Layer, and FastAPI backend. The Firestore database ensures reliable data storage, while the browser-based client interface enables seamless user interaction. The admin dashboard further provides transparency and control over the system's internal processes, enriching operational oversight.

Through the combination of AI-driven conversation, gamification, and cloud-native infrastructure, Talk2Deal illustrates a viable and scalable solution for modern e-commerce environments. The prototype not only meets the outlined objectives but also establishes a foundation for future expansion into more sophisticated negotiation and personalization systems.

10.2 Future Scope

While the current prototype delivers a functional and innovative negotiation platform, there is significant potential for extending the system into a full-fledged commercial solution. Several enhancements can be incorporated in future developments:

10.2.1 Advanced Machine Learning–Driven Pricing

The existing rule-based PricingEngine can evolve into a predictive pricing model using:

- Reinforcement learning
- Customer behavior analytics
- Historical negotiation outcomes

This would allow the system to learn optimal pricing strategies over time.

10.2.2 Multi-Product and Bulk Negotiation

Future versions can support:

- Negotiating across multiple items in a cart
- Applying bundle discounts
- Wholesale or bulk purchase negotiations

10.2.3 Personalization and User Profiling

User profiles may include:

- Purchase history
- Loyalty metrics
- Dynamic spending patterns

This allows personalized pricing or negotiation difficulty levels.

10.2.4 Voice-Based Negotiation Interface

Integrating speech-to-text and text-to-speech services would:

- Create a more natural negotiation experience
- Improve accessibility

10.2.5 Integration with Full E-Commerce Systems

The negotiation engine can be plugged into:

- Inventory systems
- Payment gateways
- Order management systems

This would complete the shopping journey beyond negotiation.

10.2.6 Advanced Gamification

Future enhancements may include:

- Leaderboards
- Tiered reward systems
- Achievement badges
- Cashback or coupon incentives based on performance

10.2.7 Analytics Dashboard for Admins

Admins can access visual analytics such as:

- Conversion rates
- Sentiment trends
- Common negotiation patterns
- Product-specific negotiation success rates

10.2.8 Multi-Language Support

Gemini can enable negotiation in:

- Regional languages
- International languages

This increases accessibility and market reach.

10.2.9 Deployment on Cloud Platforms

The system can be deployed on:

- Google Cloud Run
- AWS Lambda
- Render or Railway

Enabling global access and enterprise scalability.

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